AI 101 for Mechanical Engineers · Module 11

Loss, gradients and updates

Lesson 11.2

What you will learn

> Distinguish a loss from its gradient, separate backpropagation from the optimizer, calculate a gradient-descent update, and explain the learning rate and the limits of one update.

Lesson overview

Module 11 · Lesson 2

Loss, gradients and updates

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1 One training step, in order

Predict
$\hat{y}$
The model computes a prediction from the inputs.

→

Loss
$L(\hat{y}, y)$
One number measuring how far predictions are from the targets y.

→

Backpropagation
chain rule
Computes the derivatives of L through the layers. Changes no parameters.

→

Gradient
$\nabla_\theta L$
How L changes locally with each parameter θ.

→

Optimizer
update rule
Uses the gradient to change the parameters.

→

New parameters
θ_{new}
Used in the next step.

Keep the symbols apart: L is the loss, a single number. $\nabla_\theta L$ is its gradient, one derivative per parameter. Backpropagation calculates the gradient; the optimizer applies the update.

2 Gradient-descent update

$$\theta_{new} = \theta_{old} - \eta \nabla_\theta L$$

Supplied: $\theta_{old} = 2.0$, $\nabla_\theta L = 0.5$, $\eta = 0.1$

Adjustment: $\eta \nabla_\theta L = 0.1 \times 0.5$	**= 0.05**
$\theta_{new} = 2.0 - 0.05$	**= 1.95**

Dimensionless teaching values; the gradient is supplied, not derived. Some resources write g for $\nabla_\theta L$.

3 What the learning rate does

η too large steps can overshoot and the loss can rise

η too small progress can be very slow

η is a **training setting** chosen by the developer. Weights and biases are **fitted parameters** changed by the optimizer.

No single learning rate suits every problem.

4 What one update cannot show

One correct update does not prove that training reaches a global optimum.

It does not validate a physical machine or show that engineering limits are met.

Those claims need evidence beyond this calculation.

Key takeaway

Loss → gradients by backpropagation → update by the optimizer.

L measures the error; $\nabla_\theta L$ says how to change θ; the learning rate scales the step.

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A loss measures prediction–target discrepancy

- Training needs a numerical objective
- Loss L compares predictions with reference targets
- Defining a loss identifies the objective being fitted
- This lesson does not specify a particular loss formula

The choice of loss sets the objective

Numerical target

- Mean squared error is one option
- Averages squared prediction–target differences

Classification

- Cross-entropy losses are commonly used [4, Sec. 6.2.1.1]
- Compare predicted class probabilities with labels

> A loss value is not itself an engineering acceptance criterion.

L and ∇L give different information

Loss L

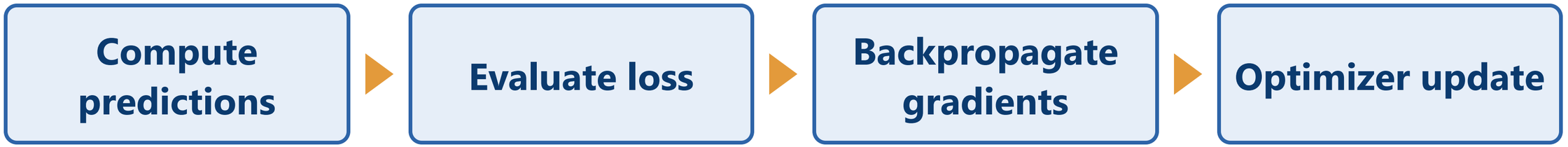
- Measures discrepancy with targets
- Defines the training objective

Gradient ∇L

- Derivatives of loss with respect to parameters
- Describes local change in the loss

> Neither is the new parameter value; the optimizer calculates that.

Four stages with separate roles



- Backpropagation uses the chain rule through the layers [4, Sec. 6.5], [61]
- The optimizer uses the gradients to change parameters
- Software may group these; their roles stay separate

Gradient descent updates a parameter

$$\theta_{new} = \theta_{old} - \eta \nabla L$$

- η is the learning rate; ∇L is the parameter's loss gradient
- Some resources write g for the gradient

The learning rate scales each update

Large learning rate

- Can overshoot

Small learning rate

- Can make progress slow

Effects depend on the problem; no universal setting. η is a training setting, not fitted.

Worked step: 2 becomes 1.95

$$\eta \nabla L = 0.1 \times 0.5 = 0.05$$
$$\theta_{new} = 2 - 0.05 = 1.95$$

- Given $\theta_{old} = 2$, gradient = 0.5, $\eta = 0.1$ (dimensionless)
- The gradient is supplied, not derived from a loss formula

One update proves little on its own

- It does not show a global optimum is reached [4]
- It does not validate a machine or its operating limits
- Forward example (a = 0.2) and this update are separate stages
- They are not one fully specified training experiment

CHECK YOUR UNDERSTANDING

Which stage computes gradients, and which stage changes the weights?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Which stage computes gradients, and which stage changes the weights?

> Backpropagation computes gradients using the chain rule. An optimizer uses the gradients to apply a parameter update.

CHECK YOUR UNDERSTANDING

Why is the change in the supplied parameter 0.05 rather than 0.5?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why is the change in the supplied parameter 0.05 rather than 0.5?

> The learning rate scales the gradient: 0.1 × 0.5 = 0.05, so the new parameter is 2 − 0.05 = 1.95.

Key takeaways

- Loss L sets the objective; ∇L describes local change
- Backpropagation calculates gradients; the optimizer applies the update
- The learning rate controls update scale; it is a setting
- One update proves neither global convergence nor equipment validation

References

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

[61] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, “Learning representations by back-propagating errors,” Nature, vol. 323, no. 6088, pp. 533–536, 1986. doi: 10.1038/323533a0

Numbers match the course reference list. The written lesson is the authoritative text.